

Plastic pollution in mangrove ecosystems: impacts, mitigation actions, and policy responses

260 sources reviewed · 1993–2026 · 7 thematic areas · 6 geographic regions

- Summary of evidence
- Thematic findings
- Evidence landscape
- Knowledge gaps
- Source table

Each thematic area has a distinct evidence profile. The cards below summarise the key findings, dominant keywords, strength of the evidence base, and geographic concentration for each area.

State of plastic pollution in mangroves

70 sources

The largest thematic category, documenting the extent and distribution of plastic waste in mangrove systems. Global syntheses now lead the evidence base alongside continued Southeast Asian fieldwork, with strong peer-reviewed backing. Key findings centre on mangroves as debris accumulation zones, microplastics in sediments, and land-based leakage pathways.

Mangrove debris accumulation

Assessment methodology

Coastal debris pathways

Land-based plastic leakage

Biodiversity

84% peer-reviewed

44% since 2023

Top region: Global (34)

Policy and governance analysis

38 sources

Dominated by intergovernmental and legal sources rather than peer-reviewed literature. Covers multilateral environmental agreements (Stockholm, Basel conventions), chemicals governance frameworks, and national plastic waste policies. Most sources are global-level analyses.

Multilateral agreements

Chemicals governance

Social impacts

POPs health effects

PBDEs

16% peer-reviewed

61% since 2023

Top region: Global (24)

Microplastics, chemical additives & POPs

28 sources

Strongly peer-reviewed evidence base examining bioaccumulation, environmental persistence, and health effects of persistent organic pollutants associated with plastics. European research is notably present. Has the oldest evidence profile — 64% published before 2023.

Bioaccumulation

POPs health effects

Environmental persistence

Fisheries contamination

Biomagnification

82% peer-reviewed

36% since 2023

Top region: Global (16)

Ecosystem services and linkages

32 sources

Examines how plastic pollution affects the ecosystem services mangroves provide — coastal protection, carbon sequestration, nursery habitat, and biodiversity. Heavy reliance on grey literature alongside peer-reviewed studies.

Mangrove ecosystem services

Social impacts

Biodiversity

Vegetation health

Mangrove

50% peer-reviewed

56% since 2023

Top region: Global (19)

Impact assessment

61 sources

Cross-cutting assessments of plastic pollution impacts on coastal ecosystems, including environmental, social, and economic dimensions. Integrates governance and scientific perspectives but lacks site-specific studies outside Asia.

Social impacts

Plastic debris

Chemicals governance

Bioaccumulation

Integrated governance

79% peer-reviewed

39% since 2023

Top region: Global (39)

Mitigation and clean-up practices

13 sources

The smallest but most recent thematic category — 77% published since 2023. Covers clean-up methodologies, bioplastic alternatives, circular economy approaches, and community-based interventions. Limited evidence for effectiveness of specific interventions.

Coastal debris pathways

Integrated governance

Ecosystem services

Coastal livelihoods

Assessment methodology

46% peer-reviewed

77% since 2023

Top region: Asia / ME (5)

Transboundary riverine plastic & chemical pollution

18 sources

A strongly peer-reviewed evidence base (83%) but almost entirely global-level studies. Examines how rivers transport plastic and chemical pollutants to coastal mangrove systems. Lacks regional case studies — a critical gap.

Chemicals governance

POPs health effects

Assessment methodology

Biodiversity

Plastic debris

83% peer-reviewed

50% since 2023

Top region: Global (12)